

# Homework 09: RNG and MC

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>\_Ex<exercise  
number>\_<FirstInitial><LastName>.py

Example: HW01\_Ex01\_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

### Exercise 1 (15 points, Python)

1. Create a 2D random walk where the random perturbation is a vector which is uniformly sampled from the unit circle (i.e., any 2D unit vector).
2. Plot the random walk for some number of points

### Exercise 2 (20 points, Python)

1. Create a 1D random walk (with your choice of distribution) which will ALWAYS stay within some bounded domain of values (say, b/t -1 and 1). You may choose any method you want for this.
2. Plot the walk as a blue line
3. Plot the upper and lower bounds as dotted red lines

### Exercise 3 (20 points, Python)

1. Randomly generate points on the surface of a cylinder
2. Create a 3D plot of these points

### Exercise 4 (25 points, Python)

1. Use MC sampling to integrate  $\sin(x)$  over  $[0, 2\pi]$  (figure out how to handle negative values so that you get an approximate answer of 0)
2. Print your result

### Exercise 5 (30 points, Python)

The exponential growth equation

$$N(t) = N_0 \exp(rt)$$

can be used to describe the growth of a population, where  $N_0$  is the initial population and  $r$  is the growth rate.

A biologist measured that a certain culture had approximately  $N_0 = 1000$  cells, but since this measurement isn't perfect, there is a standard deviation of 10. After watching the culture grow for some time, the growth rate was estimated to be  $r = 0.1d^{-1}$  (per day) with a standard deviation of 0.01.

1. Using MC sampling, obtain a distribution for the population size after 10 days, i.e.  $N(10)$
2. Plot a histogram of  $N(10)$
3. Print the mean and standard deviation of  $N(10)$